

Frontend Developer Guide

Complete integration specification for Areas, Parameters, and File Attachments System

1. Local Server & Connection

Setting	Configuration / Value
Base API URL	<code>http://127.0.0.1:8000/api</code>
Storage Asset URL	<code>http://127.0.0.1:8000/storage/</code>
Required Header	<code>Accept: application/json</code>

Backend Setup Commands:

```
# Start the backend server
cd Backend
php artisan serve

# One-time storage link creation
php artisan storage:link
```

Axios Default Header Setup

Set the header globally in your API client setup:

```
axios.defaults.headers.common['Accept'] = 'application/json';
```

2. TypeScript Data Interfaces

```
export interface FileAttachment {
  id: number;
  file_name: string;
  file_path: string;
  file_size: number;
  human_readable_size: string; // e.g. "1.45 MB", "512 B"
  mime_type: string;
  url: string; // Fully qualified URL to access/download the file
  attachable_type: 'area' | 'parameter';
  attachable_id: number;
  created_at: string; // ISO 8601 string
  updated_at: string;
}

export interface Parameter {
  id: number;
  area_id: number;
  name: string;
  details: string | null;
  files_count?: number;
  files?: FileAttachment[];
}
```

```

    area?: Area;
    created_at: string;
    updated_at: string;
  }

  export interface Area {
    id: number;
    name: string;
    description: string | null;
    parameters_count?: number;
    files_count?: number;
    parameters?: Parameter[];
    files?: FileAttachment[];
    created_at: string;
    updated_at: string;
  }

  export interface PaginationLinks {
    first: string;
    last: string;
    prev: string | null;
    next: string | null;
  }

  export interface PaginationMeta {
    current_page: number;
    from: number | null;
    last_page: number;
    path: string;
    per_page: number;
    to: number | null;
    total: number;
    max_areas?: number; // Only on /api/areas list (Always 10)
    remaining_slots?: number; // Only on /api/areas list (0 to 10)
    filters?: {
      search: string | null;
      sort_by: string;
      direction: 'asc' | 'desc';
    };
  }

  export interface PaginatedResponse<T> {
    data: T[];
    links: PaginationLinks;
    meta: PaginationMeta;
  }

  export interface SingleResponse<T> {
    data: T;
  }

```

3. Endpoints & API Reference

A. Areas (/api/areas)

GET /api/areas

Query Parameters:

- `search` (string, optional): Free-text search across area names, descriptions, parameter names, and file names.
- `sort_by` (string, optional): One of `id`, `name`, `created_at`, `updated_at`, `parameters_count`, `files_count`. (Default: `created_at`).
- `direction` (string, optional): `asc` or `desc`. (Default: `asc`).
- `per_page` (integer, optional): Number of items per page.

Response: 200 OK — `PaginatedResponse<Area>`

Note: Includes nested `files`, `parameters`, and `meta.remaining_slots`.

GET /api/areas/{areaId}

Response: 200 OK — `SingleResponse<Area>`

POST /api/areas

Content-Type: `multipart/form-data` or `application/json`

Payload:

- `name` (string, required, unique, max 255)
- `description` (string, optional, max 5000)
- `files[]` (File[], optional, up to 10 files)

Response: 201 Created | 422 Unprocessable Entity

POST `_METHOD=PUT` /api/areas/{areaId}

Content-Type: `multipart/form-data` or `application/json`

Payload: `_method`: "PUT", `name`, `description`, `files[]`

Response: 200 OK

DELETE /api/areas/{areaId}

Response: 204 No Content

Behavior: Cascades deletion to all child parameters, attachments, and storage files.

B. Parameters (/api/areas/{areaId}/parameters & /api/parameters/{parameterId})

GET /api/areas/{areaId}/parameters

Query Parameters: `search`, `sort_by` (`id`, `name`, `created_at`, `updated_at`), `direction`, `per_page`

Response: 200 OK — `PaginatedResponse<Parameter>`

POST /api/areas/{areaId}/parameters

Payload: `name` (required, unique per area), `details` (optional), `files[]` (optional)

Response: 201 Created

GET /api/parameters/{parameterId}

Response: 200 OK — `SingleResponse<Parameter>`

POST **_METHOD=PUT** /api/parameters/{parameterId}

Payload: `_method: "PUT"`, `name`, `details`, `files[]`

Response: 200 OK

DELETE /api/parameters/{parameterId}

Response: 204 No Content

C. Standalone Attachments (/api/attachments)

GET /api/attachments

Query Parameters: `attachable_type` ('area' | 'parameter'), `attachable_id`, `search`, `sort_by`, `direction`

POST /api/attachments

Content-Type: `multipart/form-data`

Payload: `attachable_type`, `attachable_id`, `files[]`

Response: 201 Created

DELETE /api/attachments/{fileId}

Response: 204 No Content

Behavior: Removes database record and unlinks file from storage disk.

4. Crucial Implementation Rules & Gotchas

1. Hard Ceiling of 10 Areas

Total areas strictly cannot exceed 10. Attempting to create an 11th area triggers a `422 Unprocessable Entity` response:

```
{
  "message": "The system limit of 10 areas has been reached. Delete an existing area before creating a new one.",
  "errors": {
    "name": [
      "The system limit of 10 areas has been reached. Delete an existing area before creating a new one."
    ]
  }
}
```

Frontend Rule: Read `response.meta.remaining_slots`. When `remaining_slots === 0`, disable the "Add Area" UI button and present an explicit indicator (e.g., `10 / 10 Areas used`).

2. Method Spoofing for File Upload Updates

PHP backends do not parse `multipart/form-data` natively when sent via HTTP `PUT` or `PATCH`.

To update an Area or Parameter with new files, execute a **POST request with `_method: 'PUT'`** in the FormData payload:

```
const formData = new FormData();
formData.append('_method', 'PUT');
formData.append('name', 'Updated Area Name');
if (selectedFiles) {
  for (let i = 0; i < selectedFiles.length; i++) {
    formData.append('files[]', selectedFiles[i]);
  }
}

await axios.post(`/api/areas/${areaId}`, formData);
```

3. Upload Restrictions Whitelist

- **Max files per request:** 10 files
- **Max size per file:** 10 MB (10,240 KB)
- **Allowed Extensions:**
 - *Images:* `.jpg`, `.jpeg`, `.png`, `.webp`, `.gif`, `.svg`
 - *Documents:* `.pdf`, `.doc`, `.docx`, `.xls`, `.xlsx`, `.ppt`, `.pptx`, `.csv`, `.txt`
 - *Archives:* `.zip`

5. Recommended UI Architecture & Screens

1. Dashboard / Area Overview

- **Top Banner:** Capacity progress meter showing active slots out of 10 (e.g. `ProgressBar: 4/10`).
- **Controls Bar:** Global search field (querying `?search=`), sort dropdown, and `+ Add Area` action button.
- **Area Cards:** Name, description, parameter count badge, attachment count badge, and action dropdown (Edit, Delete, View Detail).

2. Area Detail Screen

- **Header Section:** Area Name, Description, Edit Area trigger, Delete Area trigger.
- **Tab 1 – Parameters:** Paginated list of area parameters, "+ Add Parameter" modal (inputs: Name, Details, File Attachments), inline edit/delete triggers.
- **Tab 2 – Attachments:** File grid/table directly bound to the area, preview thumbnails for images, file extension icons, formatted sizes (via `human_readable_size`), direct download links (`href={file.url}` `target="_blank"`), drag-and-drop upload zone.